

When the comparison population changes the conclusion

Can a pathway-prioritization argument survive a change in who was measured?

Evidence

One archived A549 WT/PRDX1-KO × vehicle/AZD7648 screen. Fixed pathways; three support views; two transforms.

Result

The balance reverses mainly through the comparator. The mitochondrial arm remains nearly stable. Wider coverage includes low counts.

Scope: a finite empirical diagnosis and reproducible research record—not target validation.

The same pathway label can hide a different population

64,237 physical constructs · 16 real sample records · one source programme

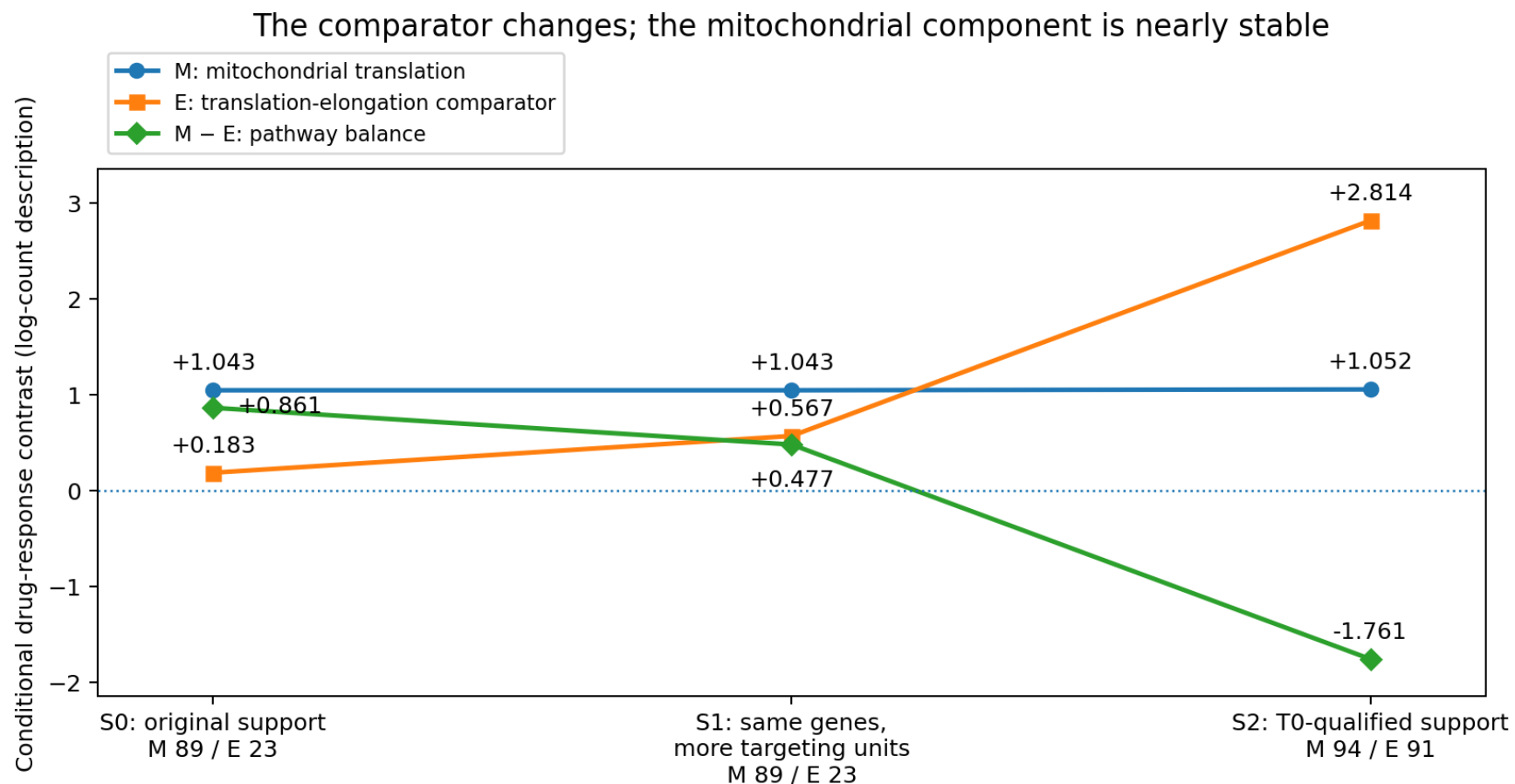
View	Genes	Targeting units	M / E
S0	Original supported	Original qualified	89 / 23
S1	Same as S0	Expanded T0 support	89 / 23
S2	All T0-supported	Expanded T0 support	94 / 91

M: Mitochondrial translation (137 directory genes)

E: Eukaryotic Translation Elongation (99 directory genes)

More covered members do not automatically mean more reliable measurements.

The comparator changes; the mitochondrial arm is stable

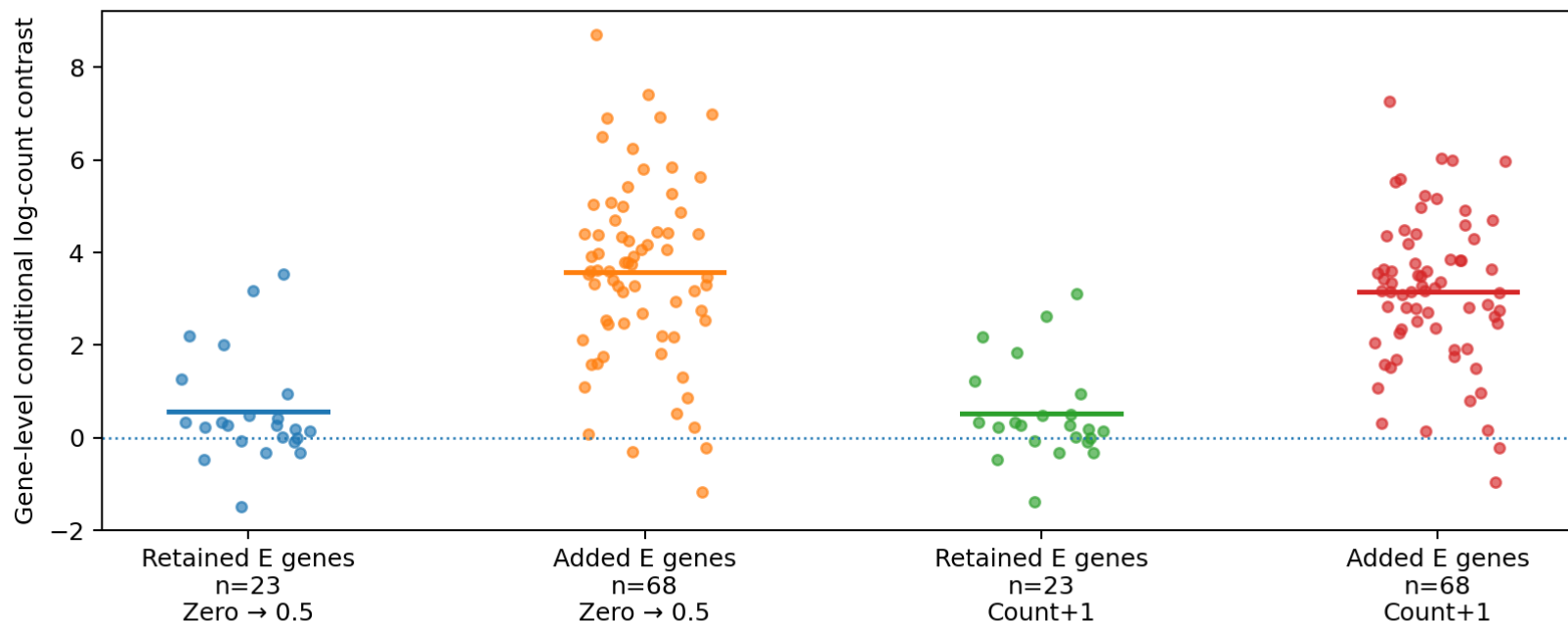


Zero-only replacement: zeros $\rightarrow$ 0.5; positive counts unchanged. No biological confidence intervals.

Different support views summarize different populations. Count+1 gives the same sign reversal (all six values in the report).

Restored comparator members drive the observed shift

The added comparator members have a different observed distribution



Zero-only means

23 retained

+0.567

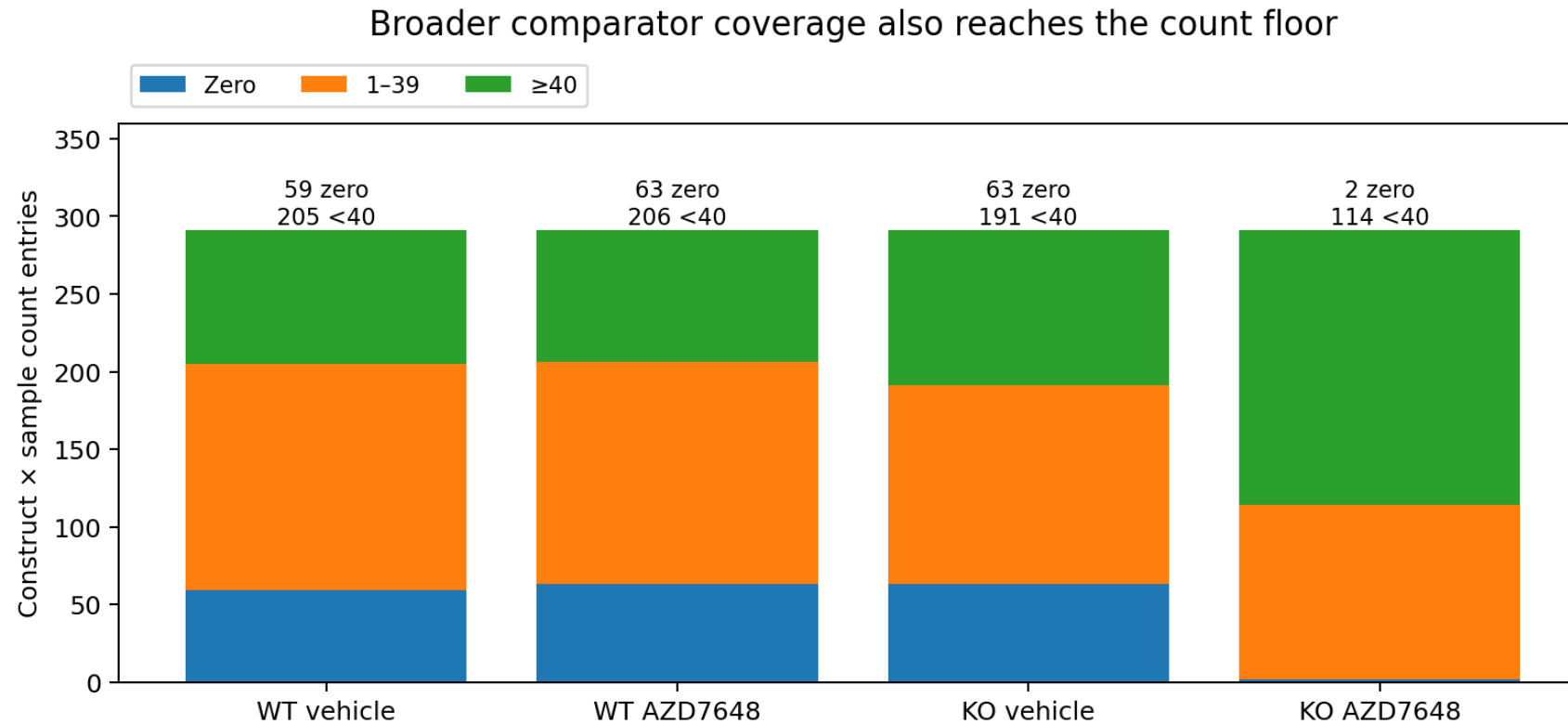
68 added

+3.574

All E genes under S2 targeting-unit support are shown. Short lines are means, not uncertainty intervals. Groups follow the frozen eligibility records. Genes are not independent biological replicates.

All members retained. The mixture is arithmetic—not a causal attribution or a new target nomination.

Coverage and count-floor risk increase together



Expanded E set: 91 genes, 97 constructs, three endpoint records per group. Each bar totals 291 entries. Categories are disjoint; the "<40" label includes zeros. Counts are not cell numbers or independent experiments.

Selection uses vehicle.

Response also subtracts vehicle.

The archive cannot fully separate composition, growth and measurement effects.

One local decision—not three layers of validation

Decision

Do not use the original positive balance as membership-insensitive support for a mitochondrial-specific target claim.

Separate histories

PRKDC: 12 of 36 source calls evaluated.
Crossing: +0.324 / +0.154; the required negative WEE1 sign failed.

Not established

A reversed mitochondrial mechanism; the opposite target; general nontransportability; an absolute survival benefit.

Contribution, attribution and responsibility

Project contribution

A local empirical comparison.

Targeting-unit / gene decomposition.

Source-table and count-export reproduction.

Explicit failed hypotheses and limits.

Original data and assistance

All experiments: cited investigators.

Standard methods: prior work.

AI: design discussion, code, tests and writing.

Personal responsibility: documented separately.

Public core: frozen transformed gene/sample values to the terminal comparison.

Source tables require separate retrieval. No FASTQ replay or new mechanism is claimed.

Backup • The exact comparison and its assumptions

$\beta(P, \text{background}) = \text{mean drug} - \text{mean vehicle}$

$\theta(P) = \beta(P, \text{KO}) - \beta(P, \text{WT})$

$\theta(\text{balance}) = \theta(M) - \theta(E)$

Common additive centering cancels; membership and count floors do not.

Two count transforms are separate sensitivity views, not independent experiments.

Culture records do not remove clone/establishment confounding.

No biological P values or confidence intervals are inferred.

Backup • Inspect the result, then follow the source

Read: `reports/DDR_TargetBridge_Case_Study.pdf`

Trace: `docs/SCIENTIFIC_CLAIM_LEDGER.json`

Scope: `docs/reproducibility.md`

Verify: `python tools/run.py verify`

Terminal comparison: `python tools/run.py core --output .build/demo`

Source tables (retrieve first): `python tools/run.py historical --output .build/history`

Data: O'Loughlin et al., Nature Chemical Biology (2026), doi:10.1038/s41589-026-02312-z. Full attribution and unchanged result tables are bundled.